



A multi-period ambulance location and allocation problem in the disaster

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Abstract

To achieve quick response in the disaster, this paper addresses the issue of ambulance location and allocation, as well as the location problem of temporary medical centers. Considering budget and capacity limitations, a multi-period mixed integer programming model is proposed and two hybrid heuristic algorithms are designed to solve this complex problem. The proposed model and algorithm are further verified in a real case study, and the numerical experiments demonstrate the effectiveness of our proposed model. Specifically, we obtain several findings based on the computational results: (1) The best locations of ambulance stations should change in each period because the demand rate changes over time. (2) Involving temporary medical centers is necessary to reduce the average waiting time of injured people. (3) It may not be optimal to allocate ambulances from the nearest ambulance stations because of potentially limited station capacity.

Keyword Ambulance station location · Ambulance allocation · Disaster response

1 Introduction

Disaster such as natural or manmade disasters may result in a massive number of injured people in a short time (Gao et al. 2019). Under these situations, a large number of seriously injured people need immediate medical care yet are located in the disaster

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areas. The role of Emergency Medical Service (EMS) in the disaster response phase is to provide out-of-hospital service to patients and transfer severe or moderate injuries from disaster areas to appropriate hospitals as soon as possible (Jánošíková et al. 2019).

When faced with the large number of people who are injured in the disaster, the current EMS resources in disaster areas may not be sufficient (Eshghi and Larson 2008). Additional resources, such as ambulances and hospitals, must be augmented in order to save as many people's lives as possible. Hence, a series of EMS decisions including the location of ambulance stations and temporary medical centers, the number of ambulances allocated to the stations and the affected areas allocated to the stations should be decided as soon as possible.

The previous studies on EMS ambulance location and allocation models such as review presented by Brothorne et al. (2003), Li et al. (2011), Farahani et al. (2012), Aringhieri et al. (2017) report that the injured people should be sent to the nearest existing hospital and do not consider building temporary medical centers or the relationship between station locations and hospital locations (Salman and Gül 2014). The performance measure was to minimize the response time or maximize the coverage to the demand area (Lam et al. 2015; Firooze et al. 2018). Most of the existing research ignores the problem of ambulance unavailability. When an ambulance is occupied with one or more patients, it may become unavailable to serve other patients. In addition, few studies in the literature consider the demand change in different periods and the ambulance location changing over time.

In this paper, we present a multi-period mixed integer programming (MIP) model to characterize the allocation issue of ambulances and the location problem of fixed facilities such as ambulance stations and temporary medical centers. The aim is to optimize ambulance location and allocation under budget and capacity limitations in the disaster. Given a set of affected areas with a differing demand rate in each period and potential locations of ambulance stations and temporary medical centers, we determine the locations of the ambulance stations and temporary medical centers while simultaneously considering budget and capacity limitations. We also allocate a limited number of ambulances to each station considering the station capacity and ambulance availability and assign the affected areas to the available stations. The objective of this model is to minimize the total patient waiting time and ambulance response time. A simulated annealing algorithm with genetic algorithm (SA–GA) and a simulated annealing algorithm with particle swarm optimization algorithm (SA–PSO) are applied to address this complex issue. Through the real case computational studies, we find (1) SA–GA shows better performance than SA–PSO, but SA–GA spends long time to reach convergence. (2) The ambulance location and allocation decision should change over the time to make the best system performance to cope with the demand change in disasters. (3) It is a life-saving measure for the potential disaster situation to prepare the potential temporary emergency medical facilities location and emergency rescue measures in the pre-disaster period. (4) When the disaster scale is relatively small, it's more important to allocate the remaining ambulances to make better system performance. When the disaster scale is large and the resource is scarce, adding more ambulances and more base stations will improve system performance effectively. (5) The ambulance dispatch strategy should be made according to the demand in the disaster since nearest-comes-first is no longer applicable in the disaster situation.

The major contributions of this study can be concluded as follows:

- (1) We propose a multi-period mixed integer programming (MIP) model to characterize the ambulance location and allocation problem considering more practical factors than existing researches.
- (2) We propose a Mixed Integer Programming (MIP) model for the problem that can be solved using heuristic algorithm. We involve the ambulance unavailability in the model by utilizing a M/M/m queuing model.
- (3) We design two heuristic algorithms (SA–GA and SA–PSO) to solve this problem and obtain several findings based on the computational results.

The rest of the paper is organized as follows: Sect. 2 reviews the literatures related to the ambulance location and allocation problem. The problem definition, assumptions and mathematical formulation are provided in Sect. 3. Section 4 presents the solution method for the proposed model. A case study and some numerical experiments are analyzed in Sects. 5 and 6. The last section concludes the paper and proposes future research focuses.

2 Literature review

There have been many studies that address the EMS ambulance station location and allocation problem. Brotcorne et al. (2003), Li et al. (2011), Firooze et al. (2018) and Aringhieri et al. (2017) provide comprehensive reviews on this subject.

In some of the early ambulance location models, the Location Set Covering Problem (LSCP) tries to cover all demand points with minimum ambulances (Toregas et al. 1971); the Maximal Covering Location Problem (MCLP) is set to maximize the total demand covered within a specified distance with limited ambulances (Church and ReVelle 1974). Both models assume that the ambulance station can always cover the demand points. Based on these two models, the double standard model (DSM) tries to maximize the total demand covered by at least two ambulances (Gendreau et al. 1997); the Maximum Expected Covering Location Problem (MEXCLP) (Daskin 1983) assumes that each ambulance is independent and each of them has the same probability of becoming unavailable. Though these two models recognize the problem that the ambulances can be unavailable, it is still not comprehensive enough. To calculate accurate busy probabilities for ambulances, a simulation model (Ahmed and Alkhamis 2009) and a queuing theory (Mendonça and Morabito 2001) are introduced into the model.

For the EMS ambulance location allocation, on the one hand, there are many studies on EMS ambulance location allocation that respond to daily incidents, and many of them have been used in practice. Yin and Mu (2012) improve the MCLP to optimally site ambulances in the state of Georgia. McCormack and Coates (2015) apply a simulation model to optimize the existing EMS vehicle fleet allocation and base station location in the London ambulance service. Liu et al. (2016) apply DSM to EMS vehicle allocation in urban areas in Chicago. Davoudpour et al. (2014) use the idea of MEXCLP and combine it with the hypercube queuing model to minimize the total traveling distance of the ambulances. This model is verified in a real case for

one of Tehran city's zones in Iran. On the other hand, there are relatively few, but a significantly increasing number of, literatures on EMS ambulance location allocation in the disaster. In a review article, Caunhye et al. (2012) mention that research on temporary response facility locations and resource allocations in the disaster do have some limitations because data may be unavailable in large-scale emergencies. There are some differences between regular EMS cases and disaster cases; for example, the demands during a disaster rise rapidly and may change over time as the disaster progresses. But in the case of daily EMS needs, the demand is usually predictable from past EMS data. For example, Maghfiroh et al. (2018) point out that the goal for the regular EMS usually is to minimize the response time between ambulance stations and demands. In contrast, in the disaster, Memari et al. (2018) point out the urgent goal is to successfully save the lives of as many injured people as possible. In addition, many regular EMS literatures focusing on ambulance assignment to increase convergence and performance measures (Kaveh and Mesgari 2019; Gai et al. 2019) ignore the locations of hospitals, but during a disaster the location of temporary medical centers may need to be considered. Few literatures such as Acuna et al. (2019) study the ambulance allocation problem considering hospital overcrowding and the heterogenous types of patients.

Some of the literature studies the ambulance location and allocation in disasters. Schneeberger et al. (2016) use DSM to deal with the location and allocation problem before a disaster and relocate the ambulance vehicles after a disaster. It combines the regular EMS and the disaster-relief EMS, however, it ignores that the demand and ambulance location may change over time after the disaster. Salman and Gül (2014) cope with the facility location and patient transportation after a disaster. This study develops a multi-period MIP model to optimize the transportation of casualties and finds the location of the temporary medical centers at the same time. The aim of the model is to minimize the total travel time and cost of opening new hospitals. Fancello et al. (2017) address the problem of allocating and reallocating vehicles involved in relief operations during major industrial disasters. The aim is to maximize the number of successfully treated injured patients. This study focuses on the vehicle allocation problem but does not consider the vehicle location during the disaster.

There are several main differences between our model and above studies. Compared with the above literatures, our studies consider the characteristics of the ambulance in the regular EMS incorporating the differences between the regular EMS and the EMS in disasters. First, we consider the ambulance unavailability as in Mendonça and Morabito (2001) using a queuing theory and we also consider the capacity of the station as in Leknes et al. (2017) and Knight et al. (2012). In the proposed model, the objective function is to minimize the total response time as in Maghfiroh et al. (2018) and we extend previous models by presenting a mixed integer programming model incorporating a M/M/m queuing model to calculate the total patients' average waiting time in the objective function.

In addition, considering the demand change over time in disasters, we propose a multi-period model that the location of the ambulance stations can displace in each period because of the demand change. Furthermore, in Leknes et al. (2017), Schneeberger et al. (2016) and Fancello et al. (2017) medical facilities are given and don't

need to build the new medical facilities. In this paper, the location decision of the temporary hospitals is also determined in the problem.

3 Problem description

In this section, an MIP ambulance location and allocation model in the disaster is presented. Given a set of affected areas with a demand rate in each period and potential locations of ambulance stations and temporary medical centers, we aim to determine the optimum locations of the ambulance stations and temporary medical centers simultaneously with budget constraints. We also allocate a limited number of ambulances to the stations considering the station capacity, allocate the ambulances to the affected areas, and assign the affected areas to the stations in each period. The objective is to minimize the total patients' waiting time and ambulance response time from the stations to the affected areas.

A multi-period planning horizon is considered because the first 72 h after a disaster is the best life-saving time in general and the highest patient demand rate usually occurs during the first 24 to 48 h after the disaster (Schultz et al. 1996). In this paper, we divide the first 72 h into 6 periods. We locate the ambulance stations and allocate the ambulances in each period considering the demand change over time. At the beginning of each period, ambulances carry medical staff from ambulance stations to the affected areas and send the injured people from affected areas to hospitals or temporary medical centers. During each period, ambulances continuously departure from hospitals and pick up injured people directly to the hospitals (Salman and Gül 2014). If there is no ambulance available, the injured people will wait in the queue. We introduce an M/M/m queuing framework in which ambulances are modeled as servers to cope with the ambulance availability (Leknes et al. 2017). With the demand arrival rate and the service rate in each affected area and the number of ambulances allocated to the affected area, the patient waiting time can be calculated by the queuing theory. The demand arrival rate in the affected area is estimated after the disaster has occurred. The service rate depends on the distances from the station to the affected area and the affected area to the available hospital.

We use an iterative framework to determine the location of the temporary medical center. Firstly, we solve the ambulance location allocation problem with the existing hospitals. Secondly, we open the temporary medical centers near the affected area with a high patient waiting time. We solve the ambulance location allocation problem and open additional temporary medical centers until the patient waiting time at all demand point are satisfactory.

Our model formulation is based on the following assumptions.

Assumption 1 The demand points are known and have been clustered as in the most literature for the EMS ambulance station location and allocation problem. Following some covering model of work of Yin and Mu (2012), every demand point can be covered by only one station.

Assumption 2 The potential points of ambulance station locations are known, and each station has its own capacity as in real case.

Assumption 3 The potential points of temporary medical centers are known.

Assumption 4 Each vehicle can carry only one injured person at a time. In the real case, only heavily and moderately injured people need treatment in hospitals. Therefore, we focus on the heavily and moderately injured people which need to be send to hospitals only.

Assumption 5 Haghani (1996) mentioned that the allocation problem must be solved with the location problem because of the capacity limit of the station. Therefore, we determine the locations of the ambulance stations and the number of ambulances in each station; we also allocate the stations to the affected areas and allocate ambulances to the affected areas at the beginning of each period.

Assumption 6 In each period, ambulances carrying medical staff from ambulance stations reach the affected areas and send the injured people to the hospitals or temporary medical centers as in real cases. If there is no ambulance available, the injured people will wait in the queue in the affected area.

Assumption 7 Injured people are sent to the nearest hospital for treatment in most literature such as Salman and Gül (2014). If the nearest hospital is full, they will be sent to the second nearest hospital.

Assumption 8 Demand rate follows an independent and identical Poisson distribution as in Memari et al. (2018). Service rate follows an exponential distribution as in Memari et al. (2018).

The multi-period ambulance location allocation model

Sets and Indices

I, i : Set and index of disaster area locations, $i \in I$.

J, j : Set and index of potential station locations, $j \in J$.

H, h : Set and index of hospital or temporary medical center locations, $h \in H$.

t : Time period, $t = 1, 2, \dots, T$.

Parameters

τ_{ij} : The average travel time from ambulance station i to ambulance station j , $i \in I, j \in J$.

σ_{ih} : The average travel time from affected area i to hospital h , $i \in I, h \in H$.

K : Total number of ambulance vehicles.

M_t : Total budget to open an ambulance station in period t , $t = 1, 2, \dots, T$.

C_j : Station capacity (the maximal number of ambulances) in station j , $j \in J$.

λ_{it} : The demand rate in the disaster area i in period t , $i \in I, t = 1, 2, \dots, T$.

μ_{it} : The service rate in the disaster area i in period t , $i \in I, t = 1, 2, \dots, T$.

v_j : The cost to open a station in j , $j \in J$.

Decision variables

$y_{jt} = 1$ if station j is open in period t ; $= 0$, otherwise, $j \in J, t = 1, 2, \dots, T$.

$o_{ijt} = 1$ if affected area i is assigned to the station j in period t ; $= 0$, otherwise, $i \in I, j \in J, t = 1, 2, \dots, T$.

z_{jt} : Number of ambulances allocated to station j in period t , $j \in J, t = 1, 2, \dots, T$.

x_{it} : Number of ambulances allocated to affected area i in period t , $i \in I, t = 1, 2, \dots, T$.

Minimize

$$f = \sum_{t=1}^T \sum_{i \in I} \left(\frac{p_{it}}{x_{it}!} \left(\frac{\lambda_{it}}{\mu_{it}} \right)^{x_{it}} \frac{x_{it} \mu_{it}}{(x_{it} \mu_{it} - \lambda_{it})^2} + \frac{1}{\mu_{it}} \right) + \sum_{t=1}^T \sum_{i \in I} \sum_{j \in J} o_{ijt} \tau_{ij}$$

Subject to

$$p_{it} = \left[\sum_{n=0}^{x_{it}-1} \frac{1}{n!} \left(\frac{\lambda_{it}}{\mu_{it}} \right)^n + \frac{1}{x_{it}!} \left(\frac{\lambda_{it}}{\mu_{it}} \right)^{x_{it}} \frac{1}{1 - \frac{\lambda_{it}}{x_{it} \mu_{it}}} \right]^{-1}, \forall i \in I, \forall t = 1, \dots, T \quad (1)$$

$$\sum_{j \in J} y_{jt} v_j \leq M_t, \forall t = 1, \dots, T \quad (2)$$

$$\sum_{i \in I} x_{it} = K, \forall t = 1, \dots, T \quad (3)$$

$$\sum_{j \in J} z_{jt} = K, \forall t = 1, \dots, T \quad (4)$$

$$\sum_{i \in I} x_{it} o_{ijt} = z_{jt}, \forall j \in J, \forall t = 1, \dots, T \quad (5)$$

$$\sum_{j \in J} o_{ijt} = 1, \forall i \in I, \forall t = 1, \dots, T \quad (6)$$

$$z_{jt} \leq C_j, \forall j \in J, \forall t = 1, \dots, T \quad (7)$$

$$z_{jt} \leq M y_{jt}, \forall j \in J, \forall t = 1, \dots, T \quad (8)$$

$$o_{ijt}, y_{jt} \in \{0, 1\}, \forall i \in I, \forall j \in J, \forall t = 1, \dots, T \quad (9)$$

$$x_{it}, z_{jt} \geq 0 \text{ and integer}, \forall i \in I, \forall j \in J, \forall t = 1, \dots, T \quad (10)$$

The objective function minimizes mean waiting time of the injured people and the response time between stations and affected areas. Constraint (1) represents the idle probability of the ambulances in affected area i in period t . Constraint (2) ensures the cost for open stations is within budget. Constraint (3) and Constraint (4) are used to ensure the flow balance of the ambulances. Constraint (3) states that the total number of ambulances allocated to the stations equals the total number of ambulances. Constraint (4) states that the total number of ambulances allocated to the affected areas equals the total number of ambulances. Constraint (5) ensures that the number of the ambulances that come from stations equals the total number of ambulances in stations. Constraint (6) ensures that one affected area can only be covered by one station. Constraint (7) ensures that the number of ambulances in the station is no more than the station capacity. Constraint (8) ensures that we can not allocate the ambulance to a station if the station is not opened. Constraint (9) and Constraint (10) define the decision variables.

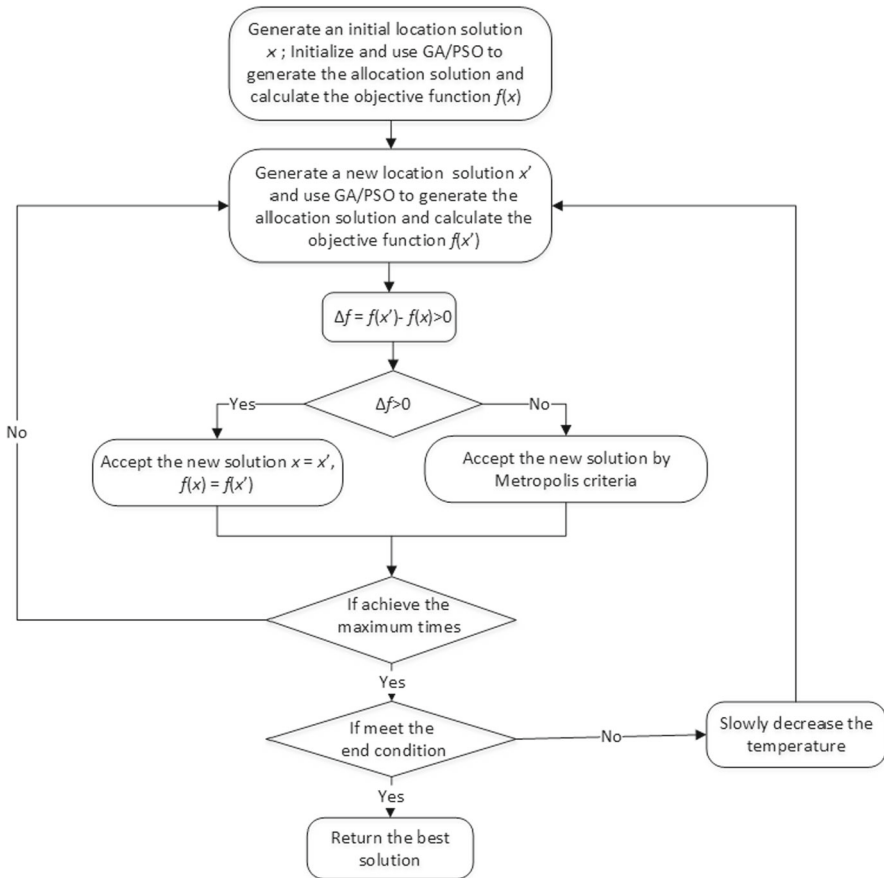


Fig. 1 Flowchart of the proposed hybrid heuristic algorithm

4 Solution method

4.1 Optimization heuristic

Since the proposed location–allocation model is a combinatorial optimization problem and proved to be NP-hard, it's hard to solve it with conventional methods in a reasonable time (Yin and Mu 2012). Therefore, we use two two-phase hybrid heuristic algorithms (SA–GA and SA–PSO) to solve this complex problem. The flowchart of the proposed hybrid heuristic algorithms is illustrated in Fig. 1.

In the first phase, we use SA to calculate the location problem. In this phase, a major decision is made: whether the potential ambulance stations should be opened. In the second phase of the algorithm, GA/PSO is implied to solve the allocation problem based on the location decision in the first phase. In the second phase, two major decisions are made: (1) the number of ambulances allocated to the affected areas; (2) whether the affected areas should be allocated to the ambulance stations. In each

iteration of SA, GA/PSO is used to find the best possible allocation solution and return the best objective function to the SA phase.

4.2 Phase 1: simulated annealing

Simulated annealing (SA) algorithm is a meta-heuristic method which imitates the physical melting process of solid materials. SA is widely used in combinatorial optimization problems and has the proven ability to find the best objective function in a short time (Zheng et al. 2013). Phase 1 describes the SA procedure used in this study.

Phase 1: Simulated annealing

```

 $T \leftarrow T_{initial}$ ;
Generate an initial solution ( $S_{Current}$ ) and compute its objective value ( $Obj_{Current}$ ),  $Obj_{Best} \leftarrow Obj_{Current}$ ,  $S_{Best} \leftarrow S_{Current}$ ;
Do
    Set  $IterNum = 0$ ;
    While ( $IterNum < MaxIterNum$ ) do
        Generate neighboring solution ( $S_{Next}$ ) and computer its objective value ( $Obj_{Next}$ );
        If ( $Obj_{Next} < Obj_{Current}$ ) Then
             $Obj_{Best} \leftarrow Obj_{Next}$ ,  $S_{Best} \leftarrow S_{Next}$ ;
        Else
            Generate a random number  $RandNum$ ;
            If ( $exp(-(Obj_{Next} - Obj_{Current})/T) > RandNum$ ) Then
                 $S_{Current} \leftarrow S_{Next}$ ,  $Obj_{Current} \leftarrow Obj_{Next}$ ;
             $IterNum \leftarrow IterNum + 1$ ;
     $T \leftarrow T * \alpha$ ;
Until termination criterion is met
Return  $S_{Best}$  and  $Obj_{Best}$ .

```

The detail description of the SA is as follows:

Initial solution For each potential ambulance station location $j \in J$, $J = \{1, 2, \dots, j, \dots, n\}$ where n is the total number of the potential ambulance station locations, the initial solution defines a proposed solution to the location of the ambulance stations. The initial solution would be a series of 0–1 variables representing whether the station should be opened or not. The possible location solution in each period can be represented by the array shown below: $Y = \{y_1, y_2, \dots, y_j, \dots, y_n\}$. Here, each y_j ($j = 1, 2, 3, \dots, n$) is a binary variable (0, 1) and n is the total number of the potential ambulance stations.

For example, consider a simple case with 5 potential ambulance station locations. One possible solution is represented as follows: {1, 0, 0, 1, 0}. The opened stations are Station 1 and Station 4.

Objective value Input the solution from SA to Phase 2 and generate the best objective value.

Neighboring solution Randomly select some positions from the potential ambulance station locations and change their states.

For example, consider 5 potential ambulance station locations and one possible solution {1, 0, 0, 1, 0}, the neighboring solution can be {0, 1, 0, 1, 0}. The opened stations are Station 2 and Station 4.

4.3 Phase 2-1: genetic algorithm

GAs have been widely used in solving difficult optimization problems. A large number of studies in the literature apply GA to solve real-word EMS ambulance location allocation problems (Shiripour and Mahdavi-Amiri 2019; Ghasemi et al. 2019). Phase 2-1 describes the GA procedure used in this study.

Phase 2-1: Genetic Algorithm
Initialize the population (P); Do Calculate the fitness values of P and find the $Individual_{Best}$ and $Fitness_{Best}$; $P_{next} \leftarrow \emptyset$; Copy elite individuals to the next generation; $P_{next} \leftarrow P_{next} + Individual_{Elite}$; While $P_{next} < popSize$ do Select individuals $parent1$, $parent2$ from P ; Generate a random number $RandNum1$; If $RandNum1 < r_x$ then Crossover the $parent1$, $parent2$ to generate offspring and add to the next generation; Mutate (P_{next}); Update the population, $P \leftarrow P_{next}$, $Individual_{Best}$ and $Fitness_{Best}$; Until termination criterion is met Return $Individual_{Best}$ and Obj_{Best} .

The detailed information of the GA algorithm is as follows:

Chromosome design For each demand area $i \in I$ and each selected ambulance station location $j \in J$, where $I = \{1, 2, \dots, i, \dots, m\}$ and $J = \{1, 2, \dots, j, \dots, n\}$, m denotes the total number of affected areas and n denotes total number of opened ambulance stations.

The chromosome, which represents the number of ambulances allocated to the affected areas and the affected areas covered by the ambulance stations in each period, is designed as an array shown below: $X = \{x_1, x_2, \dots, x_m\} \{x'_1, x'_2, \dots, x'_m\}$. The first m genes represent the number of the ambulances allocated to the affected areas. Each gene of the first m genes is an integer number between 0 and K , where K is the total number of ambulance vehicles. Eg: $x_1 = 5$ means that the number of ambulances allocated to the Affected Area 1 is 5. The last m genes represent the affected areas to be covered by the ambulance stations; each gene is an integer number between 1 and N , where N is the total number of opened ambulance stations. Eg: $x'_1 = 2$ means that the Affected Area 1 is covered by the Station 2.

Initialization Repeat to generate chromosomes until the population's size is reached.

Fitness measure The proposed objective function of the model could be measured in GA as the fitness function.

Selection Roulette wheel selection is applied to select the next population.

Crossover Two chromosomes are selected randomly from the population. A crossover point is selected randomly within the two chromosomes.

Mutation A chromosome is selected randomly from the population. A mutation point is selected randomly from the chromosome.

Infeasible solution Some operations such as initialization and mutation will produce infeasible solutions. For the total ambulance constraint, a chromosome $X = \{x_1, x_2, x_3, \dots, x_m\}$, if $x_1 + x_2 + \dots + x_m > K$, each variable in the solution will do the operation as follows and rounded: $\left(\frac{x_1}{x_1 + x_2 + \dots + x_m} * K, \frac{x_2}{x_1 + x_2 + \dots + x_m} * K, \dots, \frac{x_m}{x_1 + x_2 + \dots + x_m} * K \right)$.

Fitness measure Since the objective function is the minimum value, it is converted into the fitness function according to $fitness(x) = \frac{maxf - f(x) + \alpha}{maxf - minf + \alpha}$. Among them, $maxf$ and $minf$ are the maximum and minimum function values in the population, $f(x)$ is the current individual value, α is a random number to avoid the denominator being zero when the maximum value and the minimum value are equal.

4.4 Phase 2-2: particle swarm optimization algorithm

PSO is a parallel searching algorithm based on the intelligent behaviors of the swarm. PSO possesses prominent advantages on fast operation speed. Phase 2-2 describes the PSO procedure used in this study.

Phase 2-2: Particle Swarm Optimization Algorithm

Initialize Population

Do

For $i=1$ to Population Size

If $f(x_i) < f(p_i)$ then $p_i = x_i$, $p_g = \min(p_i)$;

For $d = 1$ to Dimension

$v_{id} = v_{id} + c_1 r_1 (p_{id} - x_{id}) + c_2 r_2 (p_{gd} - x_{id})$

$x_{id} = x_{id} + v_{id}$

if $x_{id} < a_d$ then $x_{id} = a_d$, if $x_{id} > b_d$ then $x_{id} = b_d$

End For

End For

Until termination criterion is met

The detailed information of the PSO algorithm is as follows:

x_i and v_i represent the particle's current position and velocity. p_i represent the best position of each particle. p_g represent the best position of all particle.

Particle design The particle, which has a position and velocity. Each particle's position represents the number of ambulances allocated to the affected areas and the affected areas covered by the ambulances stations, is designed the same as the GA chromosome. Each particle's velocity determines the particle's position in the next population.

Update To prevent the particles being non integer, we round off real optimum values to the nearest integer numbers.

5 Computational studies

5.1 The case description and data collection

A case study is presented to illustrate the efficiency of the mathematical model to realize quick ambulance location and allocation during the disaster. We select an area

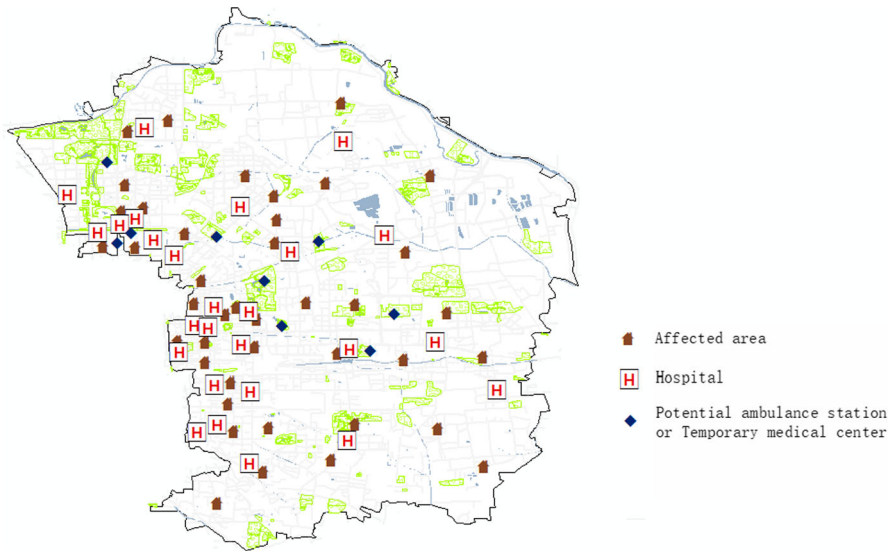


Fig. 2 The map of Chaoyang District

in Beijing, the Chaoyang district, which is the largest district in Beijing with a high risk of earthquakes. It has a population of 3,739,000 and covers an area of about 470.8 km². The map of Chaoyang district is shown in Fig. 2. In such a large area, once a disaster occurs, a series of EMS decisions need to be made immediately. It is necessary to have a quick EMS response system in place to save the as many injured people as possible. Thus our case study focuses on the respond phase of EMS in disaster situation planning. We design one earthquake scenario of magnitude 8.0 in Chaoyang district.

Time horizon The golden time of rescue is considered to be the first 72 h after a disaster (Schultz et al. 1996). The demand rate in the affected area may change over time. In response to the demand change, we divided the first 72 h into 6 time intervals. Each interval is also a medical staff shift interval and medical supplies replenishment interval.

Demand area Chaoyang District governs 24 streets and 19 districts. Because ShouDouJiChang street is far from other areas, we clustered the Chaoyang District into 42 affected areas. The population in each affected area is known from the “CHAOYANG DISTRICT BEIJING STATISTICAL YEARBOOK 2017”. According to the empirical equation of the seismic intensity attenuation model and the fragility curve, the collapse rates of buildings are set as 32.82% (Rusdy et al. 2018). Thus, total number of injuries people in each area is estimated by a certain rate of the population in each area. The demand rate in each period is presented in Table 7 in the “Appendix”, which is generated randomly within the total casualty number of each area.

Potential stations and temporary medical center locations The hospitals in Chaoyang District are selected as potential ambulance locations. Additionally, some parks and shelters are selected as potential ambulance stations and temporary medi-

cal center locations. The hospitals located in Chaoyang District are known from the listings on (<http://www.a-hospital.com/w/>) and the potential ambulance stains are got known from (<http://www.bjchy.gov.cn/chaoyang/jygy/>). There are 26 hospitals, 40 potential ambulance station locations, and 14 potential temporary medical center locations. Table 8 in the “Appendix” presents the potential ambulance station locations which contains information about current hospitals and some large-scale parks.

Travelling times The distance between two nodes are generated using Baidu Maps (a map software in China). The average vehicle velocity is set to 50 km/h. In this way, the travelling times between nodes can be calculated. In each period, the average service times T_i in each affected area which depends on average time to send a injured to the hospital can be calculated using ambulance total travelling times to send patients to the hospitals as in (Leknes et al. 2017) The average service time in the affected area contains three data points: travelling time between the hospitals and affected areas, time spent in the affected areas for attending to patients, and travelling time between the affected areas and hospitals (Leknes et al. 2017). It assumes that the average time spent in the affected area is a constant of 5 min. Hence, the average service time T_i in each affected area can be formulated as Eq. (1). Equation 1: $T_i = T_{hi} + 5 + T_{ih}$ (T_{hi} is the travel time from hospital h to affected area i , T_{ih} is the travel time from affected area i to hospital h).

Service rate The average service rate for a certain affected area is defined as how many patients that can be served per hour. It can be calculate by the service time and the ambulance number allocated to the affected area.

5.2 Results

Based on the input data, the algorithm is implemented in Matlab 2016b. The parameters of the algorithm are set as in Tables 1, 2 and 3. The results of the ambulance station location in each period are shown in Table 4. The evolution process of the solution is shown in Fig. 3.

From the results of ambulance station location in Table 4, the best location of ambulance stations is not the same in each period. The findings indicate that the ambulance station location must change with the demand change in each stage of a disaster. This finding emphasizes the importance of incorporating the different period into the model.

In Fig. 3, it is observed that SA–PSO algorithm result converges after 100 iterations and SA–GA algorithm result converges after 200 iterations. It can be seen that the SA–PSO algorithm possesses the faster convergence speed than SA–GA algorithm. But after several iterations, SA–GA algorithm shows a better performance than SA–PSO. This is probable because PSO algorithm is easy to be trapped in local optimum.

5.3 Average waiting time in the affected areas in each period

Table 5 shows the average waiting time of the injured people in 42 affected areas in each period. It is observed that the average waiting time decreases in the last two

Table 1 Parameters of SA

SA parameter	Value
Initial temperature	100
Annealing rate	0.99
Minimal temperature	90
Maximal number of generations	200

Table 2 Parameters of GA

GA parameter	Value
Population size	50
Crossover rate	0.8
Mutation rate	0.2
Maximal number of generations	1000

Table 3 Parameters of PSO

PSO parameter	Value
Population size	50
Inertia factor w	0.7
Learning factors c_1, c_2	1.4
Maximal number of generations	1000

Table 4 The results of location in each period

Period	Ambulance station					
T1	1	2	24	29	33	36
T2	1	2	6	20	33	36
T3	1	2	6	20	33	36
T4	1	2	20	29	33	36
T5	1	2	20	29	33	36
T6	1	6	20	24	26	33

periods. This implies that with the demand decrease at the end of a disaster, the average waiting time in the last two periods decreases accordingly.

The average waiting times in different affected areas vary in every period. In some affected areas, the average waiting time is especially high. This is probably because the service time is high in these affected areas.

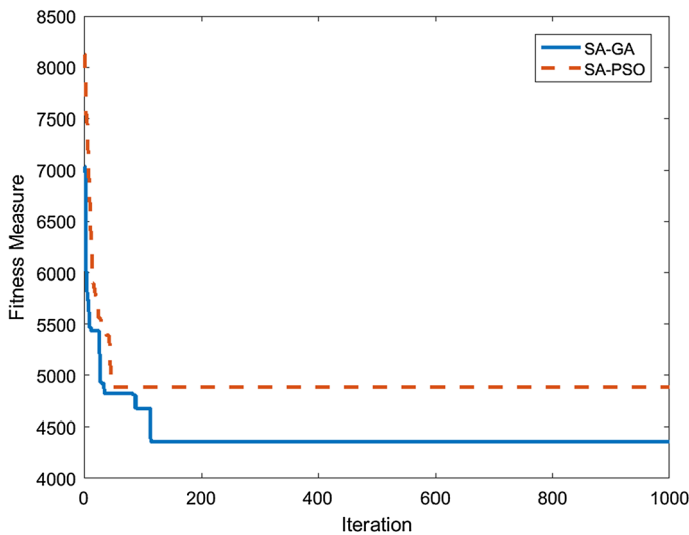


Fig. 3 Evolution process of the solutions

Table 5 The result of the waiting time in each period

Period	Average waiting time in affected areas		
	Avg.	Max.	Min.
T1	13.27	34.70	6.36
T2	13.42	34.51	6.44
T3	14.99	37.02	7.41
T4	14.94	36.65	7.87
T5	12.84	33.97	6.02
T6	12.70	33.73	6.01

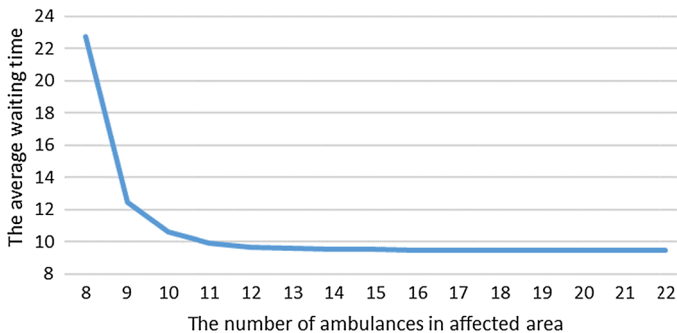
5.4 The influence of adding temporary medical centers

To decrease the high average waiting time in some affected areas, we introduce some new temporary medical centers. We use an iterative framework to determine the locations of temporary medical centers by opening temporary medical centers near the affected area with a high average waiting time constantly until the ideal result is reached.

Table 6 shows the results of adding temporary medical centers. It is observed that after adding the temporary medical centers, the average waiting time decreases dramatically in some affected areas. After adding 3 temporary medical centers, the average waiting time decreases from 13.18 to 10.42 and the maximum waiting time decreases from 34.7 to 21.28.

Table 6 The resulting waiting times after adding temporary medical centers

Adding temporary medical centers	Average waiting time in affected areas		
	Avg.	Max.	Min.
Existing hospitals	13.18	34.70	6.36
Adding station S37	11.26	27.80	5.98
Adding station S37, S47	10.66	22.45	5.95
Adding station S37, S47, S33	10.42	21.28	5.86

**Fig. 4** The influence of ambulances on the average waiting time

5.5 The influences of different parameters on the average waiting time

We now conduct some sensitivity analysis to study the impacts of different parameters on the average waiting time in the affected areas.

As shown in Fig. 4, the average waiting time in the affected areas only decreases with the number of ambulances when the ambulance number is below a threshold value. This finding implies that more ambulances do not always lead to shorter average waiting time. When the demand rate is in a certain range, continuously increasing the ambulance fleet size after it reaches a certain level will not decrease the average waiting time measurably.

Figure 5 illustrates the influence of the demand rate on the average waiting time in the affected areas. The average waiting time increases with the demand rate when the demand rate is relatively high. When the demand rate is below a certain level, the average patient's waiting time does not increase with the demand rate. This is because when the demand rate is relatively small, the number of ambulances is sufficient enough and the average patient's waiting time will not change measurably. When the demand rate exceeds a given threshold, the unavailable probability of ambulances is high and the average patient's waiting time will be long too.

We also study the relation of the ambulance station number and the objective value. The result is shown in Fig. 6. It is observed from the Fig. 6 that the objective value decreases with the number of stations. After a certain threshold, however, more ambulance stations do not improve the objective value evidently. This is because when the

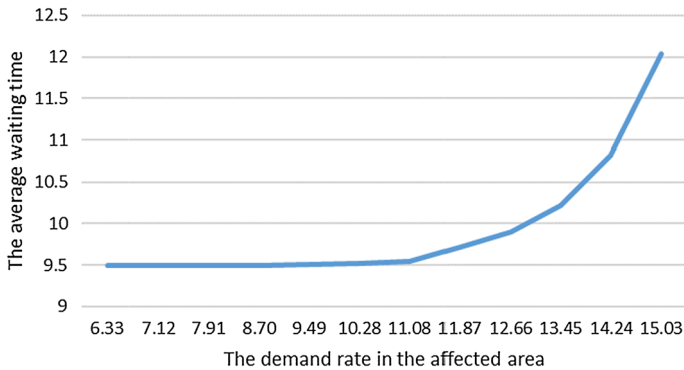


Fig. 5 The influence of demand rate on the average waiting time

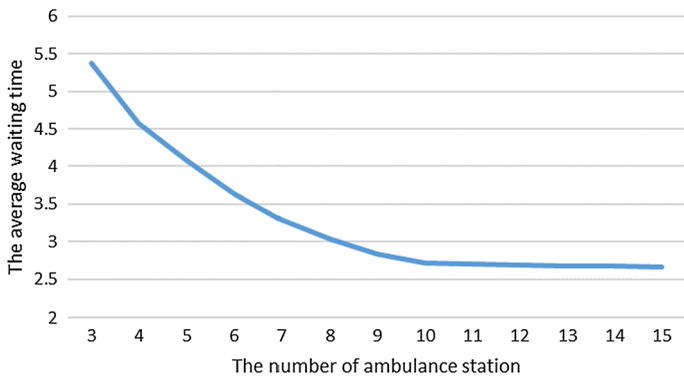


Fig. 6 The influence of number of stations on the average waiting time

number of stations is small, increasing the number of stations will decrease the distance between some affected areas and the stations. Hence, the ambulance service rate will increase, and the average patient's waiting time will decrease. But when the number of stations exceeds a threshold, adding new stations will not decrease the distance between some affected areas and the stations evidently. Therefore, there is no benefit to building excessive stations.

5.6 The capacity of the station to the result

Now we analyze the effects of station capacity to the allocation method. We solve the model under two different ambulance station capacity values. In one case, the capacity of the ambulance station is set to 1000 (a sufficiently large number). In another case, the capacity of the ambulances station is fixed as 250.

Figures 7 and 8 show the result of the ambulance allocation. From Fig. 8, it is observed that because of the limited capacity of the ambulance station, the ambulances in the affected areas do not always come from the nearest ambulance station. Only

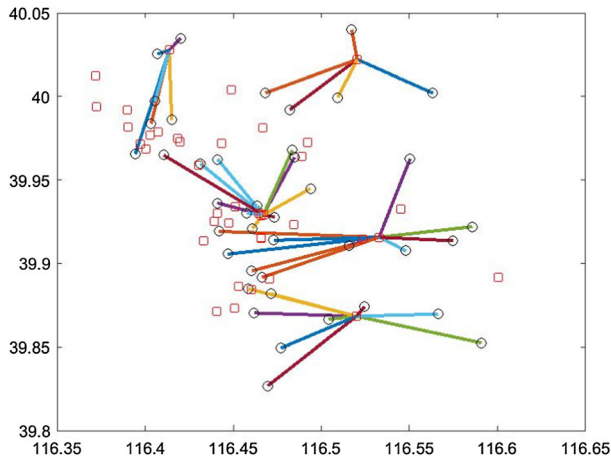


Fig. 7 The ambulance allocation with station capacity (capacity = 1000)

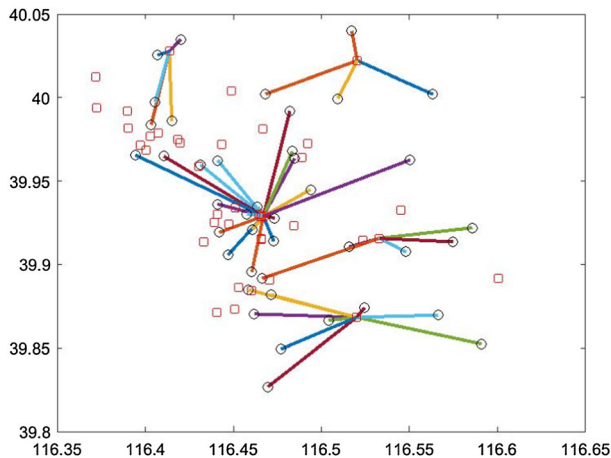


Fig. 8 The ambulance allocation without station capacity (capacity = 250)

with a sufficiently large capacity of the ambulance station do the ambulances allocated to the affected areas come from the nearest ambulance station (as shown in Fig. 7).

When considering the capacity of the station, it's obvious that allocating the ambulances from the nearest station does not always lead to performance improvements. Because of the capacity limitation, ambulances may not come from the nearest station.

5.7 Managerial implications

We summarize some managerial implications from the above experiments:

- (1) The best locations and allocations of ambulance change from period to period because the demand rate changes. Thus, it is necessary to change the ambulance

location and allocation decision during disasters to make the best system performance.

- (2) It is necessary to build temporary medical centers for disaster relief. Adding temporary medical centers can decrease the average injured people waiting time measurably and add additional hospital capacity for the injured people. For decision makers, it is vital to determine the potential temporary emergency medical facilities location and prepare some emergency medical arrangements in the pre-disaster period. It is a time saving procedure to improve the survival rate for the wounded under the disaster.
- (3) Providing more ambulances does not necessarily lead to shorter injured people waiting time; when the number of ambulances is increased beyond a certain level, the average waiting time will maintain a certain value. For the decision maker, it's more important to allocate the remaining ambulances to make better system performance when the disaster scale is relatively small. When the disaster scale is large and the current recourse is scarce, adding more ambulances will improve system performance effectively.
- (4) Increasing the number of ambulance base stations does not always lead to better performance. When the disaster scale is large, adding more base station will improve system performance effectively. Since more base stations requires additional funding and the budget for disaster relief is limited, it is more beneficial to balance the number of ambulance stations and ambulances to make the best system performance in the limited budget.
- (5) It is common to allocate ambulances to the affected areas from the nearest ambulance stations. But with capacity limitations, it may not be the optimal solution to allocate the ambulance from the nearest ambulance station. In the disaster situation, the ambulance dispatch strategy that nearest-comes-first is no longer applicable and a reasonable ambulances dispatch strategy need to reassign allocation for the specific disaster situation.

6 Conclusions

In this paper, we present an optimization model to address the issue of ambulance location in the disaster planning horizon. The major contribution to the literature is the combination of the temporary medical center location and ambulance location and allocation problem. The problem contains some practical factors to be more realistic than earlier problems, such as the time-dependent changes of demand, and the location of temporary medical centers. The average waiting time of the wounded is incorporated in the objective function by utilizing a M/M/m queuing model to calculate the ambulance unavailability.

A multi-period MIP model is proposed and two hybrid heuristic algorithms are designed to solve this model. We applied our model to the case of a disaster in the Chaoyang district in Beijing. An earthquake disaster scenario in which demand rate changes over time is designed. Some numerical experiments are conducted to identify managerial implications on the ambulance location and allocation problems in the disaster. The computational results show that to cope with the demand change in

disasters, the ambulance location and allocation decision should change over the time to make the best system performance. For decision makers, it is a life-saving measure for the potential disaster situation to prepare the potential temporary emergency medical facilities and emergency rescue measures in the pre-disaster period. When the disaster scale is relatively small, it's more important to allocate the remaining ambulances to make better system performance. When the disaster scale is large and the current resource is scarce, adding more ambulances and more base station will improve system performance effectively. Since the budget is limited, it's better to balance the number of ambulance stations and ambulances to make the best system performance in the limited budget. The ambulance dispatch strategy should be made according to the demand in the disaster since the nearest-comes-first is no longer applicable in the disaster situation.

Although our study achieves a decision approach that combines the location of ambulance stations, the allocation of ambulances to demand, and the location of temporary medical centers on the EMS planning in the disaster, there are some potential limitations of our study. In the model, we assume that the patients are sent to the nearest hospital and will be sent to the second nearest hospital if the nearest hospital is full. The exact capacity in hospital are not considered in the model. In addition, we assume that there are only one type of the ambulances. And in the real case, there may exist more than one type of the ambulance. In the real disaster situation, the capacity of hospital should be considered and heterogeneous ambulances should be considered too. Thus, in the future research, models may be more comprehensive and more realistic. The heterogeneous patients and different performance measures would be incorporated to the model. Also, the capacity of the hospital will be considered too. Furthermore, time-dependent travel times due to the road damage would be incorporated to the model. The equity to allocate the ambulances to the affected areas would be analyzed in the experiments too.

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Appendix

See Tables 7 and 8.

Table 7 Demand rate in affected area in each period

Affected area	Demand rate (person/h)						Total (person)
	T1	T2	T3	T4	T5	T6	
A1:Jianwai	67	134	157	114	96	47	7380
A2:Chaowai	140	163	119	100	70	48	7680
A3:Hujialou	108	154	182	215	251	75	11,820
A4:Sanlitun	44	91	148	128	108	64	6996
A5:Zuojiashuang	76	110	156	185	218	255	12,000
A6:Xiangheyuan	40	57	81	97	114	133	6264
A7:Hepingjie	212	251	295	345	149	103	16,260
A8:Anzhaen	112	133	156	182	79	54	8592
A9:Yayuncun	57	82	117	138	162	190	8952
A10:Xiaoguan	195	167	142	120	84	58	9192
A11:Jiuxianqiao	81	197	232	272	166	117	12,780
A12:Maizidian	86	74	63	53	37	26	4068
A13:Tuanjieshu	41	60	85	101	119	139	6540
A14:Liulitun	104	148	242	206	175	72	11,364
A15:Balizhuang	100	205	286	335	243	144	15,756
A16:Shuangjing	86	176	287	245	208	124	13,512
A17:Jinsong	88	127	180	214	252	294	13,860
A18:Panjiayuan	94	135	192	314	268	228	14,772
A19:Fatou	137	117	100	84	59	41	6456
A20:Datum	304	260	221	186	131	91	14,316
A21:Wangjing	103	149	212	251	296	346	16,284
A22:Aoyuncun	246	210	178	151	106	73	11,568
A23:Donghu	139	119	101	85	60	41	6540
A24:Nanmofang	71	103	239	204	173	146	11,232
A25:Gaobeidian	198	169	143	121	85	59	9300
A26:Jiangtai	106	91	77	65	46	31	4992
A27:Taiyanggong	64	93	215	184	156	132	10,128
A28:Xiaohongmen	35	50	116	99	84	71	5460
A29:Shibalidian	47	68	157	134	114	97	7404
A30:Pingfang	146	125	106	90	63	43	6876
A31:Dongfeng	129	110	94	79	56	38	6072
A32:Laiguangying	229	195	166	141	99	68	10,776
A33:Changjing	52	74	87	103	120	36	5664
A34:Sanjianfang	264	226	192	162	114	79	12,444
A35:Guanzhuang	203	103	71	146	173	238	11,208
A36:Jinshan	80	68	33	48	94	110	5196

Table 7 continued

Affected area	Demand rate (person/h)						Total (person)
	T1	T2	T3	T4	T5	T6	
A37:Sunhe	68	80	94	110	33	48	4020
A38:Cuigezhuang	59	70	86	73	41	29	4524
A39:Dongba	177	151	128	109	76	53	8328
A40:Heizhuanghu	78	67	57	48	34	23	3684
A41:Dougezhuang	16	23	33	39	46	53	2520
A42:Wangsiying	77	66	56	47	33	23	3624

Table 8 The potential ambulances stations

No.	Ambulance station location
S1	Beijing Obstetrics and Gynecology Hospital, Capital Medical University (East)
S2	Beijing Obstetrics and Gynecology Hospital, Capital Medical University (West)
S3	Beijing Chaoyang Hospital, Capital Medical University
S4	Wangjing Hospital, China Academy of Chinese Medical Sciences
S5	China-Japan Friendship Hospital of the Ministry of Health
S6	People's Liberation Army No. 306 Hospital
S7	Beijing Anzhen Hospital, Capital Medical University
S8	Beijing First Hospital of Integrated Traditional Chinese and Western Medicine
S9	Beijing Ditan Hospital, Capital Medical University
S10	Chinese Academy of Medical Sciences Cancer Hospital
S11	Children's Hospital of the Capital Institute of Pediatrics
S12	Beijing Dangyangliu Hospital
S13	Civil Aviation General Hospital
S14	China Medical University Aviation General Hospital
S15	Chinese People's Armed Police Force Beijing Corps Hospital
S16	Beijing Huaxin Hospital
S17	Coal General Hospital
S18	The Third Affiliated Hospital of Beijing University of Chinese Medicine
S19	Beijing Chaoyang District Second Hospital
S20	Shuangqiao Hospital, Chaoyang District, Beijing
S21	Beijing Coking Chemical Factory Hospital
S22	Chaoyang District Chinese Medicine Hospital, Beijing
S23	Beijing Chaoyang District Maternal and Child Health Hospital
S24	Beijing Tibetan Hospital
S25	Beijing Wuzhou Women and Children Hospital

Table 8 continued

No.	Ambulance station location
S26	Shuangqiao Hospital, Chaoyang District, Beijing
S27	North Xiaohe Park North
S28	Olympic Forest Park South Park West
S29	Yuan Dadu Chenghuang Ruins Park East
S30	Anzhen Yongxi Park
S31	Sun Palace Park West
S32	West of the Taiba River Green Belt
S33	North West of Chaoyang Park
S34	Red Scarf Park West
S35	Southwest of Beijing Liyuan
S36	Northeast of Xinglong Park
S37	National Sport Stadium
S38	Olympic Sports Center Stadium
S39	Chaoyang Stadium
S40	Workers Stadium

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